The $\mathfrak{sl}_n$ Cactus Representation Theorem: Interval Reversals, Crystal Structures, and a Corrected Statement

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Abstract

We prove that the interval reversal operators on $V^{\otimes k}$ ($V = \mathbb{C}^n$, arbitrary $n \geq 2$), constructed from the swap operator P (the ice-rule R-matrix at $q = 0$), define a representation of the cactus group J_k. The proof is dimension-free: it depends only on the fact that P is the permutation matrix on $V \otimes V$, not on the dimension of V. This extends the $\mathfrak{sl}_2$ result to all $\mathfrak{sl}_n$.

We also analyse the relationship between this representation and the Henriques–Kamnitzer crystal commutor, finding that they are *distinct*: the reversal permutation does not commute with Kashiwara crystal operators on a single tensor product. The crystal commutor for identical factors $B(\omega_1)^{\otimes k}$ is trivial (the identity), while the reversal is nontrivial. The reversal does, however, intertwine the standard and reordered tensor product crystal structures—a tautological consequence of the signature rule being defined relative to a reading order.

1 The Theorem

Theorem 1 ($\mathfrak{sl}_n$ Cactus Representation)**.** *Let $n \geq 2$, $V = \mathbb{C}^n$ with standard basis $\{e_1, \ldots, e_n\}$, and $P \in \mathrm{End}(V \otimes V)$ the swap operator: $P(e_a \otimes e_b) = e_b \otimes e_a$. This is the six-vertex ice-rule R-matrix at $q = 0$ for $\mathfrak{sl}_n$.*

For $k \geq 2$ and $1 \leq i < j \leq k$, define the commutor $\sigma_{[i,j]} \in \mathrm{End}(V^{\otimes k})$ as the operator that reverses the order of tensor factors i through j, acting as the identity on all other factors. Then:

1. ***Cactus representation.*** *The map $\rho\colon J_k \to \mathrm{GL}(V^{\otimes k})$ defined by $s_{[i,j]} \mapsto \sigma_{[i,j]}$ is a group homomorphism.*

2. ***Factorisation through S_k.*** *The representation factors as $\rho = \rho_{S_k} \circ \phi$, where $\phi\colon J_k \twoheadrightarrow S_k$ is the natural surjection and ρ_{S_k} is the permutation representation on tensor factors.*

3. ***Dimension-free.*** *The proof uses only the symmetric group structure, not $\dim V$. It holds identically for all $n \geq 2$.*

2 Proof of Theorem ??

The proof is identical to the $\mathfrak{sl}_2$ case (see our earlier proof of the Cactus Representation Theorem, April 10, 2026). We reproduce it here for completeness, noting that no step uses $\dim V = 2$.

2.1 Construction

Definition 2. *The commutor* $\sigma_{[i,j]} \in \mathrm{End}(V^{\otimes k})$ *is the image of the longest element* $w_0 \in S_{j-i+1}$ *under the permutation representation* $S_{j-i+1} \to \mathrm{GL}(V^{\otimes(j-i+1)})$, $s_r \mapsto P_{i+r-1}$, *embedded into* $\mathrm{End}(V^{\otimes k})$. *Explicitly:*

$$\sigma_{[i,j]}|b_1 \cdots b_k\rangle = |b_1 \cdots b_{i-1}\, b_j\, b_{j-1} \cdots b_{i+1}\, b_i\, b_{j+1} \cdots b_k\rangle,$$

where $b_r \in \{1, \ldots, n\}$ *for all* r.

2.2 Verification of cactus relations

Relation (C1): Involutivity

$\sigma_{[i,j]}$ is the image of $w_0 \in S_{j-i+1}$ under a group homomorphism. Since $w_0^2 = e$ in the symmetric group, we have $\sigma_{[i,j]}^2 = I$. Alternatively: reversing a substring twice restores the original order. □

Relation (C2): Disjoint commutativity

If $[i,j] \cap [k,l] = \emptyset$, then $\sigma_{[i,j]}$ and $\sigma_{[k,l]}$ act on disjoint sets of tensor positions. Products of operators P_r with $r \in [i, j-1]$ commute with products of P_r with $r \in [k, l-1]$, since $P_r P_s = P_s P_r$ for $|r-s| \geq 2$. □

Relation (C3): Nesting

Proposition 3. *If* $[k,l] \subseteq [i,j]$, *then* $\sigma_{[i,j]}\, \sigma_{[k,l]} = \sigma_{[i+j-l,\, i+j-k]}\, \sigma_{[i,j]}$.

Proof. Identify positions $\{i, \ldots, j\}$ with $\{1, \ldots, m\}$ via $t \mapsto t - i + 1$, where $m = j - i + 1$. Then $\sigma_{[i,j]}$ corresponds to $w_0 \in S_m$ (the map $t \mapsto m + 1 - t$), and $\sigma_{[k,l]}$ to $w_0^{[k',l']}$ (reversal on $[k', l']$ with $k' = k - i + 1$, $l' = l - i + 1$).

The nesting relation becomes $w_0\, w_0^{[k',l']}\, w_0^{-1} = w_0^{[m+1-l',\, m+1-k']}$, which we verify by computing the action on an arbitrary position t:

Case 1: $t \notin [m+1-l',\, m+1-k']$. Then $w_0(t) = m + 1 - t \notin [k', l']$, so $w_0^{[k',l']}$ fixes it, giving $\tau(t) = w_0^{-1}(w_0(t)) = t$.

Case 2: $t \in [m+1-l',\, m+1-k']$. Write $t = m + 1 - s$ with $s \in [k', l']$. Then $\tau(t) = w_0^{-1}(k' + l' - s) = m + 1 - k' - l' + s = (m + 1 - k') + (m + 1 - l') - t$, which is the reversal within $[m+1-l', m+1-k']$. □

Remark 4. This proof uses *only* the symmetric group — the fact that $\sigma_{[i,j]}$ permutes tensor factor positions. It never references the dimension of V, the crystal structure, or any Lie-theoretic data. The same proof works verbatim for $V = \mathbb{C}^n$ (any $n \geq 1$), and indeed for any vector space V.

3 Relationship to the Henriques–Kamnitzer Crystal Commutor

The original conjecture (PROVE.md) asserted that the reversal representation ρ coincides with the Henriques–Kamnitzer crystal commutor on $B(\omega_1)^{\otimes k}$ and intertwines the crystal operators $\tilde{e}_\alpha$, $\tilde{f}_\alpha$. We show that both claims are **false**, and identify the correct statement.

3.1 The HK commutor on identical crystals is trivial

Proposition 5. *Let $B = B(\omega_1)$ be the fundamental crystal of $\mathfrak{sl}_n$. The Henriques–Kamnitzer crystal commutor σ_{HK} on $B^{\otimes k}$ acts as the identity.*

Proof. The HK commutor $\sigma_{B_1,B_2}\colon B_1 \otimes B_2 \to B_2 \otimes B_1$ is the unique crystal isomorphism between the two tensor product crystals. When $B_1 = B_2 = B$, the objects $B_1 \otimes B_2$ and $B_2 \otimes B_1$ are *equal* as crystals — the Kashiwara tensor product rule depends on the crystal data $(\varphi_i, \varepsilon_i)$ of each element, not on which "copy" it belongs to.

The tensor product $B(\omega_1) \otimes B(\omega_1)$ decomposes as $B(2\omega_1) \oplus B(\omega_2)$ for $n \geq 3$ (and $B(2\omega_1) \oplus B(0)$ for $n = 2$), each component with multiplicity 1. The unique crystal automorphism of a connected crystal $B(\lambda)$ is the identity (it is determined by the image of the highest weight vector, which must be the unique highest weight vector). Therefore σ_{HK} is the identity on each component, hence the identity on $B \otimes B$.

By the coherence condition of coboundary categories, the commutor for composite objects is built from pairwise commutors:

$$\sigma_{B^{\otimes p},B^{\otimes q}} = (\sigma_{B,B^{\otimes q}} \otimes \mathrm{id}_{B^{\otimes(p-1)}}) \circ \cdots$$

When each pairwise commutor is the identity, all composite commutors are the identity. □

3.2 The reversal is not a crystal morphism

Counterexample 6. *For $\mathfrak{sl}_2$, $B = B(\omega_1) = \{1, 2\}$ with $\tilde{f}_1(1) = 2$. The tensor product $B \otimes B$ has crystal structure (Kashiwara rule, left factor has priority when $\varphi_1(\text{left}) > \varepsilon_1(\text{right})$):*

$$\tilde{f}_1(1,1) = (2,1), \quad \tilde{f}_1(2,1) = (2,2), \quad \tilde{f}_1(1,2) = 0.$$

The swap $P\colon (a,b) \mapsto (b,a)$ fails to commute with $\tilde{f}_1$:

$$P(\tilde{f}_1(1,1)) = P(2,1) = (1,2), \qquad \tilde{f}_1(P(1,1)) = \tilde{f}_1(1,1) = (2,1).$$

*Since $(1,2) \neq (2,1)$, the swap P is **not** a crystal endomorphism of $B \otimes B$. The same failure occurs for all $n \geq 2$.*

Corollary 7. *The reversal $\sigma_{[i,j]}$ does not commute with the Kashiwara crystal operators $\tilde{f}_\alpha$ on $(B(\omega_1)^{\otimes k}, \Delta_{\mathrm{std}})$.*

3.3 What the reversal does intertwine

The reversal $\sigma_{[i,j]}$ is not a crystal endomorphism, but it is a crystal *isomorphism between different crystal structures* on the same underlying set.

Definition 8. *For a reading order $\pi = (\pi_1, \ldots, \pi_k)$ (a permutation of $\{1, \ldots, k\}$), define the π-crystal structure on $\{1, \ldots, n\}^k$ by the Kashiwara tensor product rule with the α_i-signature read in the order $\pi_1, \pi_2, \ldots, \pi_k$ instead of $1, 2, \ldots, k$.*

Proposition 9. *Let $\pi_{[p,q]}$ denote the reading order obtained from the standard order $(1, \ldots, k)$ by reversing positions p through q. Then the reversal $\sigma_{[p,q]}$ is a crystal isomorphism*

$$\sigma_{[p,q]}\colon (B(\omega_1)^{\otimes k}, \Delta_{\mathrm{std}}) \xrightarrow{\sim} (B(\omega_1)^{\otimes k}, \Delta_{\pi_{[p,q]}}).$$

Proof. The Kashiwara tensor product rule determines $\tilde{f}_i(w)$ from the α_i-signature of the word $w = (w_1, \ldots, w_k)$, which is a string of $+$ and $-$ symbols at positions where $w_j = i$ and $w_j = i+1$ respectively, read in a specified order. The reduced signature (after cancelling $+-$ pairs) determines which position j_0 is modified by $\tilde{f}_i$ (the leftmost uncancelled $+$).

The reversal $\sigma_{[p,q]}$ simultaneously:

1. permutes the values in positions $p, \ldots, q$, and
2. corresponds to reading those positions in the reversed order.

Since the signature symbols are determined by the *values* at each position, and the reduction algorithm depends on the *order* in which they are read, applying $\sigma_{[p,q]}$ to the values while reversing the reading order produces exactly the same reduced signature (with position labels relabeled). Therefore

$$\sigma_{[p,q]} \circ \tilde{f}_i^{\Delta_{\mathrm{std}}} = \tilde{f}_i^{\Delta_{\pi_{[p,q]}}} \circ \sigma_{[p,q]}.$$

□

Remark 10. This intertwining is essentially tautological: permuting both the values *and* the reading order preserves the reduced signature and hence the crystal action. It does not give a crystal endomorphism (which would require preserving a *single* crystal structure).

3.4 The two representations compared

Property	Reversal $\sigma_{[i,j]}$	HK commutor σ_{HK}
Action on $B^{\otimes k}$	Nontrivial (reversal)	Trivial (identity)
Crystal endomorphism?	No	Yes (trivially)
Satisfies cactus relations?	Yes	Yes (trivially)
Source	R-matrix at $q = 0$	Coboundary category
Intertwines	$\Delta_{\mathrm{std}} \leftrightarrow \Delta_\pi$	$\Delta_{\mathrm{std}} \to \Delta_{\mathrm{std}}$

Remark 11. The HK commutor becomes nontrivial for tensor products of *different* crystals: $\sigma_{B(\lambda),B(\mu)}$ with $\lambda \neq \mu$ is the combinatorial R-matrix, which is a nontrivial crystal isomorphism between $B(\lambda) \otimes B(\mu)$ and $B(\mu) \otimes B(\lambda)$ (genuinely different crystal structures). The triviality for identical crystals is a consequence of $B \otimes B = B \otimes B$, not a deficiency of the theory.

4 Computational Verification

4.1 Cactus relations

All three cactus relations (C1)–(C3) were verified by explicit permutation matrix computation for the following cases:

$\mathfrak{g}$	$\dim V$	k	$\dim V^{\otimes k}$	Generators	Result
$\mathfrak{sl}_2$	2	3, 4	8, 16	3, 6	ALL PASS
$\mathfrak{sl}_3$	3	3, 4	27, 81	3, 6	ALL PASS
$\mathfrak{sl}_4$	4	3, 4	64, 256	3, 6	ALL PASS

4.2 Crystal morphism failure

The swap $P\colon (a,b) \mapsto (b,a)$ was verified to *not* commute with crystal operators $\tilde{f}_i$ for $\mathfrak{sl}_n$, $n = 2,3,4$.

For $\mathfrak{sl}_2$, $k = 3$: all 5 non-zero crystal operator applications fail to commute with the reversal.

For $\mathfrak{sl}_3$, $k = 3$: 22 out of 28 non-zero applications fail (the 6 that "pass" do so by accident, not by structure).

4.3 Crystal intertwining

The intertwining property $\sigma_{[p,q]} \circ \tilde{f}_i^{\Delta_{\mathrm{std}}} = \tilde{f}_i^{\Delta_{\pi_{[p,q]}}} \circ \sigma_{[p,q]}$ was verified for all intervals $[p,q]$ with $\mathfrak{sl}_2$ ($k = 3,4$) and $\mathfrak{sl}_3$ ($k = 3,4$). All tests pass.

5 Discussion

5.1 What the theorem establishes

The $\mathfrak{sl}_n$ Cactus Representation Theorem confirms that the algebraic structure of interval reversals in the quasi-solvable regime ($q = 0$) is *universal*: it is the cactus group J_k, regardless of the rank n of the Lie algebra. This extends the categorical phase diagram from $\mathfrak{sl}_2$ to all $\mathfrak{sl}_n$:

Regime	Category	Structure group	Holds for
Solvable ($q \neq 0$)	Braided monoidal	B_k	all $\mathfrak{sl}_n$
Quasi-solvable ($q = 0$)	Coboundary monoidal	J_k	all $\mathfrak{sl}_n$
Symmetric ($q = 0, u = 0$)	Symmetric monoidal	S_k	all $\mathfrak{sl}_n$

5.2 What the theorem does not establish

The reversal representation is a *group representation*, not a *crystal morphism*. It acts on the vector space $V^{\otimes k}$ by permuting basis vectors, but it does not preserve the crystal structure defined by the Kashiwara operators. The crystal commutor of Henriques–Kamnitzer acts trivially on $B(\omega_1)^{\otimes k}$ when all factors are identical.

This suggests that the correct setting for the coboundary monoidal structure at $q = 0$ is the category of *representations* (where the swap is the categorical braiding/commutor), not the category of *crystals* (where the commutor for identical objects is trivial). This is consistent with the Alqady–Stroinski result that the Temperley–Lieb category TL_0 (a category of representations, not crystals) is coboundary monoidal.

5.3 Corrected relationship to Henriques–Kamnitzer

The reversal permutation $\sigma_{[i,j]}$ and the HK crystal commutor are related but distinct:

- Both define cactus group representations on $B(\omega_1)^{\otimes k}$.
- The HK commutor is the identity (trivial representation).
- The reversal is the image of J_k in S_k acting by permuting tensor factor positions.
- The reversal intertwines different tensor product crystal structures (standard $\leftrightarrow$ reordered).

- The HK commutor is a crystal endomorphism (identity) of a single crystal structure.

The meaningful connection to HK arises for tensor products of *different* crystals $B(\lambda_1) \otimes \cdots \otimes B(\lambda_k)$, where the HK commutor is nontrivial and the combinatorial R-matrix provides the crystal isomorphism. The passage from the quantum group R-matrix to the combinatorial R-matrix under $q \to 0$ is the crystal-theoretic content of the braided $\to$ coboundary transition.

Acknowledgements

Computational verification performed using NumPy and a custom implementation of the Kashiwara tensor product rule (signature method).

References

[1] A. Henriques and J. Kamnitzer, *Crystals and coboundary categories*, Duke Math. J. **132** (2006), no. 2, 191–216.

[2] R. Alqady and M. Stroinski, *On the coboundary monoidal structure of the Temperley–Lieb category at $q = 0$*, preprint.

[3] M. Kashiwara, *On crystal bases of the q-analogue of universal enveloping algebras*, Duke Math. J. **63** (1991), no. 2, 465–516.

[4] V. Buciumas and T. Scrimshaw, *Quasi-solvable lattice models and Demazure atoms*, preprint.